

**National Workshop
On
Familiarization of Indian Ocean Land Atmosphere (IOLA)
Mesoscale Coupled Model to Advancing Severe Weather and
Ocean Hazard Prediction
Dates: 5-9 January 2026
Venue: Swosti Premium Hotel, Jaydev Vihar, Bhubaneswar
Organized by NIT Rourkela**

Day 1 (5 Jan 2026)	
9:30 – 12:00 Inaugural session	
9:30 – 9:35 am	Welcoming the guests to the workshop by Ms. Imamah Ali
9:35 – 9:40 am	Lighting of lamps
9:40 – 9:45 am	Welcoming the gathering by Prof. Krishna K Osuri , Convener
9:45 – 10:00 am	Introducing the theme of the workshop by Dr. S G Gopalakrishnan PI, MM3 project
10:00 – 10:15 am	Remarks by the Guest of Honor, Prof. U C Mohanty Advisor, MM3 project
10:15 – 10:30 am	Remarks by the Guest of Honor Dr. V S Prasad Director/Head, National Center for Medium Range Weather Forecast (NCMRWF)
10:30 – 10:45 am	Address by the Chief Guest, Dr. M Mohapatra Director General of Meteorology, India Meteorological Department
10:45 – 11:00 am	Presidential Address and Felicitation of the Guests by Prof. K. Umamaheshwar Rao , Director, NIT Rourkela, Patron of the workshop
11:15 – 11:30 am	Vote of thanks by Dr. Naresh Krishna Vissa , Head, Department of Earth and Atmospheric Sciences, NIT Rourkela
11:30 am – 12:00 pm	High Tea
12:00 – 1:00 pm	Talk by Dr. V S Prasad – Modelling efforts at NCMRWF
1:00 – 2:00 pm	Break for Lunch
2:00 – 2:45 pm	Prof. UC Mohanty: Introduction to Monsoon rainfall Prediction and challenges
3:00 – 3:45 pm	Talk by S G Gopalakrishnan (Multi-nesting capabilities and model Predictability)
4:00 – 4:45 pm	Krishna K Osuri - Land Surface Processes and Numerical modelling of severe weather
5:00 – 6:00 pm	Talks from participants (15 min each)*

Day 2 (6 Jan 2026)	
9:30 – 1:00 Session: Monsoon Heavy rainfall and model predictability	
9:00 – 9:45 am	Model dynamics – by S G Gopalakrishnan
10:00 – 10:45 am	Dr. Ananda Kumar Das, IMD Delhi: Intense Rainfall Prediction – Challenges to address
11:00 – 11:15 am	High Tea
11:15 am – 12:00 pm	Dr. Ashish Routray NCMRWF - Severe weather prediction - capabilities of global and regional models
12:15 – 1:00 pm	Prof. Rajiv Bhatla, BHU Varanasi – Model predictability of Monsoon rainfall at various scales
1:15 – 2:00 pm	Break for Lunch
	Session: Familiarization with IOLA
2:00 – 5:00 pm	Introduction to IOLA (workflow, libraries, model namelist explanation, etc) – by Krishna, Rajit, Gopal, Sasanka, and team
	3.30 – 3:45 PM (Break)
5:00 – 6:00 pm	Talks from participants (15 min each)*

Day 3 (7 Jan 2026)	
9:30 – 1:00 Session: Severe Convective activity and numerical modelling	
9:00 – 9:45 am	Prof. A P Dimri, Director, IIG – Himalayan Weather and Challenges
10:00 – 10:45 am	Dr. Anupam Hazra, IITM - Severe weather and microphysics
11:00 – 11:15 am	High Tea
11:15 am – 12:00 pm	Dr. Gyan Mohapatra CSIR4PI – Microphysics in high resolution models
12:15 – 1:00 pm	Dr. KC Gowda CSIR4PI – Integration of physics-informed AI/ML methods to improve severe weather prediction
1:15 – 2:00 pm	Break for Lunch
	Session: Familiarization with IOLA
2:00 – 5:00 pm	Introduction to IOLA (installation of libraries, model components, and Atmospheric model run) – by Krishna, Rajit, Gopal, Sansaka, and team
	3.30 – 3:45 PM (Break)
5:00 – 6:00 pm	Talks from participants (15 min each)*

Day 4 (8 Jan 2026)	
9:30 – 1:00 Session: Ocean processes and coupling with the Atmospheric model	
9:00 – 9:45 am	Dr. Sudheer Joseph, INCOIS – Ocean coupled models and coastal processes
10:00 – 10:45 am	Prof. Sourav Sil, IIT BBS – Regional Ocean Modelling
11:00 – 11:15 am	High Tea
11:15 am – 12:00 pm	Dr. Ananda Kumar Das, IMD – Impact-based forecast – A forecaster perspective from coupled models
12:15 – 1:00 pm	Dr. B R R Hari Prasad, NCMRWF - Understanding severe weather from regional models-Predictability issues
1:15 – 2:00 pm	Break for Lunch
	Session: Familiarization with IOLA
2:00 – 5:00 pm	Coupled version set up and running procedure – by Sasanka, Rajith, Krishna, Gopal, and team
	3.30 – 3:45 PM (Break)
5:00 – 6:00 pm	Talks from participants (15 min each)*

Day 5 (9 Jan 2026)	
9:00 – 10:30 Session: AI/ML Applications and coupled modelling for severe weather & Operational Oceanography	
9:00 – 9:45 am	Prof. Dev Niyogi, UT Austin and PI of MM3 project - Digital Twins, AI, and Extreme Weather with IOLA 2.0 (online)
9:45 – 10:30 am	Dr. Ankur Kumar, NASA-Impact Team (online) – Incorporation of AI/ML methods to improve convection and lightning
10:30 – 11:00 am	High Tea
11:00 am – 1:00 pm: Concluding session – Way forward to improve the IOLA model	
11:00 am – 12:00 pm	Feedback from IOLA model team, Resource persons from IMD, INCOIS, IITM, NCMRWF, and Participants
12:00 – 12:15 pm	Dr. G.K.Patra, Director, CSIR4PI, Bangalore (Guest of honour)
12:15 – 12:30 pm	Dr. A Suryachandra Rao, Director, IITM Pune (Guest of honour)
12:30 – 12:45 pm	Dr. M Ravichandran, Secretary, MoES, Delhi (Chief Guest)
12:45 – 12:55 pm	Felicitation to the guests by the Dean, NIT Rourkela
12:55 – 1:00 pm	Vote of thanks by the Convener

** Exact titles of the talks will be updated soon.

*** List of the participants who will be presenting the talk**

06/01/2026 5:00 – 6:00 pm	Dr. Sahadat Sarkar Mr. G Sripathi Ms. Angela Sabu
07/01/2026 5:00 – 6:00 pm	Dr. Prabodha Pradhan Dr. Hara Prasad Nayak Mr. Kesireddy Lakshman
08/01/2026 5:00 – 6:00 pm	Mr. K Srinivasa Rao Mr. Narayan Reddy

Please mail your presentation prior to your talk at: nitriolaworkshop@gmail.com

